

# MoMem: Decoupling Factual and Experiential Memory for Lifelong LLM Agents

Anonymous ACL submission

## Abstract

Lifelong LLM agents face heterogeneous memory demands. Some interactions require the agent to preserve explicit facts such as user preferences and environment states, while others require it to internalize transferable strategies for reasoning, planning, and tool use. Existing agent memory systems usually rely on a single memory substrate, either external textual memory or parametric adaptation, making it difficult to support factual precision and experiential reuse well within the same system. Importantly, experiential memory does not map one-to-one to implicit memory: workflow-like experience can often remain in explicit memory, while reasoning-dense experience is better internalized. We propose **Mixture of Memory (MoMem)**, a hybrid memory architecture that decouples factual memory from experiential memory across storage, invocation, and update. MoMem maintains factual memory and explicitizable procedural traces in an explicit memory module, while using an implicit memory module implemented with LoRA adapters and a Logic-Indexed Trajectory Bank for hard-to-externalize experiential knowledge. At inference time, a memory router selects which explicit memories to inject and whether implicit memory should be activated. After each interaction, a memory decoupler separates different memory signals and writes them back through different mechanisms. Our goal is not to argue that a hybrid system should dominate specialized single-substrate methods on every isolated benchmark.

## 1 Introduction

Large language model (LLM) agents are increasingly expected to operate over long horizons, interact across tasks, and improve through continued use. In such settings, memory is not only a mechanism for extending context length, but also the basis for preserving prior knowledge, adapting to users, and accumulating experience over time. Realis-

tic agent workloads place heterogeneous demands on memory: some tasks require accurate retention of explicit facts such as user preferences, entity attributes, and environment states, while others require reuse of strategies for reasoning, planning, coding, and tool use. Following [Hu et al. \(2025\)](#), we refer to these two functional roles as *factual memory* and *experiential memory*. This creates a central challenge for memory design: the same architecture must preserve factual accuracy while also supporting the accumulation and reuse of experience over time.

Most existing agent memory architectures, however, are still organized around a single memory substrate, making it difficult to support these two functions equally well. A major line of work stores memory outside the agent in textual form, often referred to as *explicit memory*, or more broadly *external memory* ([Zhang et al., 2025b](#); [Huang et al., 2026](#)). These systems perform strongly on factual recall, preference tracking, and long-horizon consistency ([Zhong et al., 2024](#); [Packer et al., 2023](#); [Chhikara et al., 2025](#); [Xu et al., 2025](#); [Li et al., 2025](#); [Wang et al., 2024b](#); [Zhang et al., 2025a](#); [Fang et al., 2025](#)), and can also support the part of experiential memory that is representable as workflows or structured traces ([Wang et al., 2024b](#); [Zhang et al., 2025a](#); [Fang et al., 2025](#)). Yet they remain weaker on reasoning-intensive tasks such as mathematics, where useful signals are often logically rather than semantically related ([Liu et al., 2025](#)). When mismatched information is injected into long contexts, it can become noise rather than useful reasoning support ([Liu et al., 2024](#); [Zhou et al., 2025](#); [Bui, 2026](#)). More broadly, prompt-level reuse remains closer to retrieving prior traces than to dynamically accumulating reusable experience ([Wei et al., 2025](#)).

A different direction stores memory in model parameters, often described as *implicit memory* or *internal memory* ([Zhang et al., 2025b](#); [Huang et al.,](#)

2026). This substrate is better suited to reasoning-dense experience that is difficult to externalize as compact text, because it can internalize reusable behavior rather than replay prior traces through context (Yuksekgonul et al., 2026; Zuo et al., 2025; Wei et al., 2025). But it is a less reliable substrate for newly emerging facts. Prior work shows that retrieval is often preferable to fine-tuning for factual injection (Ovadia et al., 2024; de Luis Bala-guer et al., 2024), while factual edits in parameter space remain constrained by the trade-off among reliability, generalization, and locality (Wang et al., 2024a). Under repeated updates, this difficulty is further amplified by interference between edits, especially for sparse long-tail knowledge (Nguyen et al., 2026; Yi et al., 2025).

More importantly, the difficulty is not only that no single substrate is equally well suited to both functions. Forcing factual memory and experiential memory into a single substrate also couples how memory is invoked at inference time and how it is updated after interaction. In mixed workloads, explicit retrieval can return memories in which factual details and strategy traces are entangled, bloating the context and introducing low-relevance noise (Liu et al., 2024; Zhou et al., 2025; Bui, 2026). For parametric memory, if the same parameter space must absorb both factual details and transferable experience, updates become more frequent and more interfering, increasing cost and catastrophic forgetting (Kirkpatrick et al., 2017; Wang et al., 2024a; Luo et al., 2023). Taken together, these observations suggest that the problem is not only one of representational mismatch, but also one of interaction-level interference between different memory functions.

We address these challenges with **Mixture of Memory (MoMem)**, a hybrid memory architecture designed to support factual and experiential memory in a more balanced manner under heterogeneous workloads. MoMem follows three coordinated design principles: it assigns memories by representational fit, keeping factual knowledge and explicitizable procedural experience in explicit memory while reserving implicit memory for hard-to-externalize reasoning experience; it introduces a *memory router* that decides which explicit memories should be injected and whether implicit memory should be activated for the current query; and it introduces a *memory decoupler* that separates post-interaction memory signals so that different information is written back through different mech-

anisms.

Our contributions are as follows:

- We frame agent memory in long-horizon, mixed-task settings as a *balance problem*: the goal is not to maximize one memory function in isolation, but to support factual and experiential memory well within the same system while reducing interference between them.
- We propose **MoMem**, a hybrid memory architecture that combines heterogeneous substrates with routing and decoupling, so that factual information, explicitizable procedural traces, and hard-to-externalize reasoning experience are stored, invoked, and updated differently.
- We evaluate MoMem on factual-only, implicit-experience-only, and pairwise mixed factual-implicit-experience settings, and show that it achieves stronger cross-task balance and smaller degradation under mixed workloads than single-substrate baselines.
- We further perform ablations on the router and decoupler, showing that MoMem’s gains do not come from hybridization alone, but from explicitly separating how heterogeneous memory signals are selected and updated.

## 2 Related Work

Recent surveys suggest two complementary ways to organize agent memory. Hu et al. (2025) distinguish memory by *function*, emphasizing factual memory and experiential memory as two different roles in long-horizon agents. In parallel, Huang et al. (2026) and Zhang et al. (2025b) distinguish memory by *substrate*, separating memory maintained outside the model from memory absorbed into model parameters. We follow these two views throughout this paper: the former motivates the functional requirements of agent memory, while the latter clarifies the strengths and limitations of existing implementations.

### 2.1 Explicit Memory for LLM Agents

The most established class of agent memory is *explicit* or *external* memory, which stores prior interactions outside the model and retrieves them at inference time (Huang et al., 2026; Zhang et al., 2025b). Representative systems include MemoryBank (Zhong et al., 2024), MemGPT (Packer

et al., 2023), Mem0 (Chhikara et al., 2025), A-Mem (Xu et al., 2025), and CAM (Li et al., 2025). Although these methods differ in how memory is organized—for example as dialogue summaries, user facts, agentic notes, or structured records—they share the same basic paradigm of external storage plus retrieval-augmented reuse. This makes them especially well suited to factual recall, preference tracking, and long-horizon consistency, where interpretability, editability, and precise overwriting are important.

Recent work has also pushed explicit memory beyond plain fact storage by converting trajectories into more reusable procedural forms. Agent Workflow Memory (Wang et al., 2024b), G-Memory (Zhang et al., 2025a), and Memp (Fang et al., 2025) all show that workflows, hierarchical traces, and procedural abstractions can improve the reuse of explicit experience. Even so, this family still depends on retrieval and prompt injection, so its effectiveness remains bounded by the quality of retrieved context, especially when semantic similarity is not a reliable proxy for logical usefulness.

## 2.2 Implicit Memory and Parametric Adaptation

*Implicit* or *internal* memory instead stores knowledge in model parameters or parameter-efficient modules (Huang et al., 2026; Zhang et al., 2025b). Recent examples include Evo-Memory (Wei et al., 2025), Learning to Discover at Test Time (Yuksekgonul et al., 2026), and TTRL (Zuo et al., 2025). Rather than replaying prior traces in the prompt, these methods adapt the model itself so that useful behavior can be reused as latent reasoning patterns or policy improvements. This makes them a natural fit for the part of experiential memory that is difficult to externalize as compact text.

At the same time, prior work also shows that parametric memory is a weaker substrate for precise factual maintenance. Comparisons between retrieval and fine-tuning find that factual knowledge injection through parameter updates is often less reliable than external retrieval (Ovadia et al., 2024; de Luis Balaguer et al., 2024). Work on lifelong model editing and repeated knowledge updates further highlights tensions among reliability, locality, generalization, and stability in parameter space (Wang et al., 2024a; Nguyen et al., 2026; Yi et al., 2025). These results suggest that internal memory is better aligned with reusable experience than with high-precision factual storage.

## 2.3 Trade-offs of Single-Substrate Memory

The literature above points to a broader limitation of *single-substrate* memory. On the explicit side, more retrieved context does not necessarily translate into better use of memory: long-context studies show that language models do not use relevant information uniformly across context positions (Liu et al., 2024), while reasoning benchmarks with growing context complexity show systematic degradation as irrelevant structure increases (Zhou et al., 2025). In retrieval settings, HopRAG (Liu et al., 2025) further shows that semantic similarity alone can be insufficient when the task depends on multi-step reasoning.

On the implicit side, continual updating in a shared parameter space raises stability and forgetting problems. Classical continual learning work identifies catastrophic forgetting as a fundamental challenge in sequential parameter updates (Kirkpatrick et al., 2017), and recent studies show that this issue remains substantial for LLMs under continual fine-tuning and repeated editing (Wang et al., 2024a; Luo et al., 2023). Taken together, direct comparisons between explicit and implicit memory (Zhang et al., 2025b) and between retrieval and fine-tuning (Ovadia et al., 2024; de Luis Balaguer et al., 2024) point to the same conclusion: existing approaches expose a persistent trade-off between factual precision and experiential reuse. Our work builds on this tension, but focuses on a different question: how to design a memory architecture that remains balanced when these two demands are repeatedly interleaved within the same long-horizon workload.

## 3 Method

### 3.1 Overview

Figure 1 illustrates the overall architecture of MoMem. Given a user request, MoMem first retrieves textual candidates from the explicit memory module. For implicit memory, MoMem does not retrieve trajectories by raw semantic similarity. Instead, it first performs a lightweight pre-reasoning to judge whether the request exhibits nontrivial reasoning structure and is therefore likely to benefit from implicit memory. Only in that case does it derive a structured logic sketch of the methods and rough reasoning steps likely to be needed, and retrieve logic-matched trajectory evidence from the Logic-Indexed Trajectory Bank. These retrieved results are passed to the memory router, which selects

the explicit memories to inject into the prompt and makes the final decision about whether the current request should activate the implicit memory pathway. The agent then generates a response using the routed memory configuration. After the interaction is completed, MoMem sends the full trajectory and feedback to the memory decoupler, which extracts verbalizable and internalizable memory signals and writes them back through the corresponding memory modules.

The two control points in this process are the router and the decoupler. The router operates before generation. It does not simply append all retrieved information to the context; instead, it filters explicit memories according to the current request and uses logic-indexed trajectory evidence to judge whether implicit memory is likely to help. The decoupler operates after generation. It determines which parts of the trajectory can be normalized into explicit records and which parts should instead be internalized as reusable reasoning behavior.

### 3.2 Explicit Memory Module

The explicit memory module stores information whose reusable value can be stably externalized as a compact textual record. It is therefore the substrate for memory that should remain verbalizable, inspectable, and editable, rather than being absorbed into model parameters. MoMem does not store raw trajectories in this module. Instead, it stores distilled textual memory items extracted from interaction and normalized into standalone records that can be semantically retrieved and directly reused at inference time.

Formally, the explicit memory bank is defined as

$$\mathcal{M}_t^{\text{ext}} = \{m_i^{\text{ext}}\}_{i=1}^{N_t^{\text{ext}}}, \quad (1)$$

where each item is represented as

$$m_i^{\text{ext}} = (c_i, \mu_i). \quad (2)$$

Here,  $c_i$  denotes a distilled textual record to be reused, and  $\mu_i$  stores metadata such as time. The content field contains the normalized memory item extracted from prior interaction, while the metadata field records auxiliary information needed for retrieval and management. Thus, the explicit module provides a controllable substrate for memory that must remain accessible as an external record.

### 3.3 Implicit Memory Module

The implicit memory module stores reusable reasoning experience whose value is not reliably cap-

tured as a compact textual record. It consists of parameter weights  $\Delta_t$  and a Logic-Indexed Trajectory Bank  $\mathcal{P}_t$ . The parameter weights internalize transferable reasoning behavior, while the bank provides retrieval support for deciding when this internalized experience should be activated.

Rather than storing trajectories for prompt reuse, the bank stores a compact logic descriptor extracted from each accepted trajectory. This allows MoMem to retrieve past experience by reasoning structure rather than by surface semantic similarity. In this sense,  $\mathcal{P}_t$  is not a second explicit memory store for task solving; it is an index that helps identify when implicit memory is likely to be useful.

Each entry in the Logic-Indexed Trajectory Bank is represented as

$$p_i = (x_i, \ell_i, \mu_i), \quad \ell_i = (m_i, s_i), \quad (3)$$

where  $x_i$  is the original request,  $m_i$  denotes the methods used in the trajectory,  $s_i$  denotes a rough sketch of the reasoning steps, and  $\mu_i$  stores metadata such as time. Here,  $m_i$  records which method or methods were used, while  $s_i$  preserves the coarse order of the reasoning steps without full detail. For a mathematics trajectory,  $m_i$  could indicate methods such as reduction to a standard quadratic form or substitution, while  $s_i$  could summarize “derive the unknown quantity, substitute into the governing equation, verify the candidate.” Together,  $\ell_i$  describes the reusable reasoning structure of the trajectory at a compact level.

### 3.4 Inference-Time Memory Routing

The memory router controls how the two memory substrates participate in generation. Given a new request  $x_t$ , MoMem first performs a lightweight pre-reasoning step:

$$q_t^R = \mathcal{A}_{\text{LLM}}(x_t), \quad (4)$$

where  $q_t^R$  is either a structured logic sketch aligned with the bank descriptor  $\ell_i = (m_i, s_i)$ , or  $\emptyset$  when the current request does not exhibit enough reasoning structure to warrant logic-oriented retrieval. When produced,  $q_t^R$  summarizes the methods and rough reasoning steps required by the current request. This step does not solve the task itself; it only produces a retrieval-oriented abstraction when such an abstraction is useful.

MoMem then gathers two different kinds of evi-



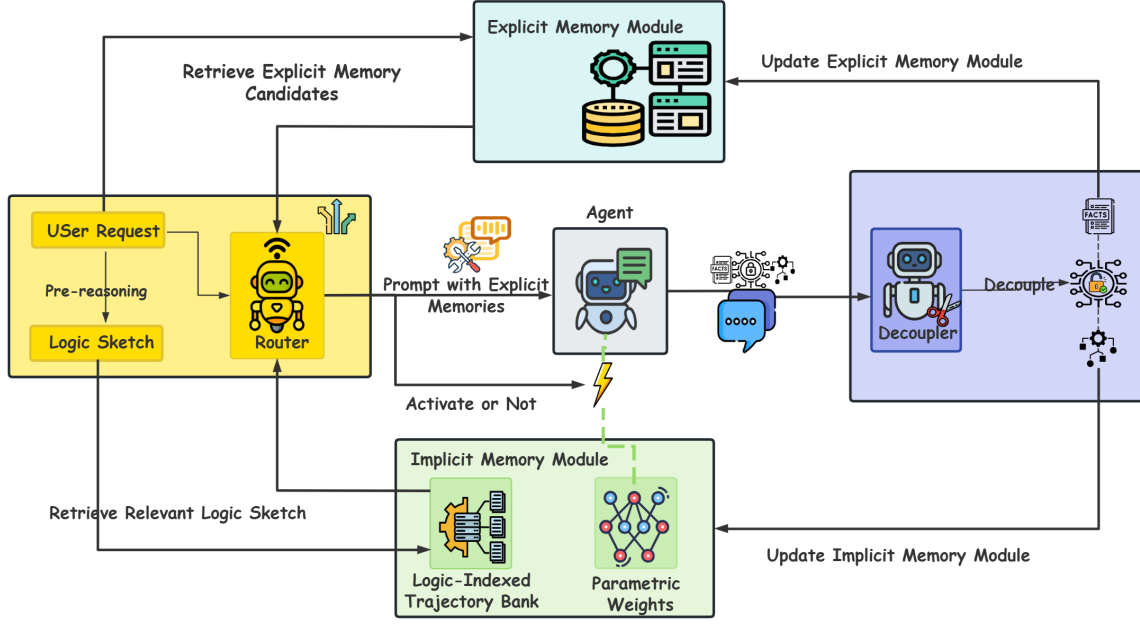


Figure 1: Overview of MoMem. Left: during inference, the memory router retrieves verbalized records from the explicit memory module and logic-matched trajectory evidence from the Logic-Indexed Trajectory Bank, then decides which explicit records to inject and whether implicit memory should be activated. The bank outputs routing evidence rather than prompt content. Right: after interaction, the memory decoupler separates the completed trajectory into verbalizable records and internalizable reasoning experience, then writes them back through different memory pathways.

dence:

$$\begin{aligned} \mathcal{C}_t^{\text{ext}} &= \text{Ret}_{\text{ext}}(x_t, \mathcal{M}_t^{\text{ext}}), \\ \mathcal{C}_t^P &= \begin{cases} \text{Ret}_P^{\text{logic}}(q_t^R, \mathcal{P}_t), & q_t^R \neq \emptyset, \\ \emptyset, & q_t^R = \emptyset. \end{cases} \end{aligned} \quad (5)$$

Here,  $\text{Ret}_{\text{ext}}$  retrieves explicit-memory candidates through semantic retrieval over the extracted textual records in  $\mathcal{M}_t^{\text{ext}}$ . In contrast,  $\text{Ret}_P^{\text{logic}}$  is invoked only when a non-empty logic sketch is produced. In that case it matches  $q_t^R$  with the stored logic descriptors  $\ell_i = (m_i, s_i)$  in  $\mathcal{P}_t$ , where  $m_i$  records methods and  $s_i$  records rough reasoning steps, so the resulting set  $\mathcal{C}_t^P$  reflects reasoning-structure similarity rather than raw semantic similarity between requests.

These two candidate sets play different roles:  $\mathcal{C}_t^{\text{ext}}$  provides possible prompt content, while  $\mathcal{C}_t^P$  provides routing evidence for implicit-memory activation rather than content to be directly injected into the prompt. The pre-reasoning is only a permissive filter; the final activation decision is still made by the router. MoMem therefore uses an LLM-based router  $\mathcal{R}_{\text{LLM}}$  to jointly decide which explicit memories to keep and whether implicit

memory should be activated:

$$(\tilde{\mathcal{C}}_t^{\text{ext}}, z_t) = \mathcal{R}_{\text{LLM}}(x_t, \mathcal{C}_t^{\text{ext}}, \mathcal{C}_t^P), \quad (6)$$

where  $\tilde{\mathcal{C}}_t^{\text{ext}} \subseteq \mathcal{C}_t^{\text{ext}}$  is the selected explicit-memory subset and  $z_t \in \{\text{SKIP}, \text{ACTIVATE}\}$  is the router’s activation decision for the implicit memory module.

After routing, the generation context is constructed as

$$\text{Prompt}_t = [x_t; \tilde{\mathcal{C}}_t^{\text{ext}}], \quad (7)$$

and the response is generated by

$$y_t \sim \begin{cases} \pi_{\theta}(\cdot | \text{Prompt}_t), & z_t = \text{SKIP}, \\ \pi_{\theta, \Delta_t}(\cdot | \text{Prompt}_t), & z_t = \text{ACTIVATE}. \end{cases} \quad (8)$$

This design prevents MoMem from indiscriminately injecting all retrieved memories into the prompt, thereby reducing context clutter and retrieval noise, while still allowing historical trajectories to influence inference through reasoning-structure matching. The trajectory bank thus affects generation through routing, rather than by acting as a second prompt-side memory source.

### 3.5 Post-Interaction Memory Decoupling

If routing determines how memory is used during inference, decoupling determines how memory is formed after interaction. Rather than writing the completed interaction into a unified memory pool, MoMem applies an LLM-based decoupler  $\mathcal{D}_{\text{LLM}}$  to separate explicit-side content from implicit-side reasoning experience:

$$(E_t^{\text{ext}}, d_t^{\text{ext}}, E_t^{\text{int}}, d_t^{\text{int}}) = \mathcal{D}_{\text{LLM}}(\tau_t, u_t), \quad (9)$$

where  $E_t^{\text{ext}}$  denotes verbalizable memory-bearing content to be written to the explicit module, while  $E_t^{\text{int}}$  denotes internalizable reasoning experience for the implicit module. The variables  $d_t^{\text{ext}}$  and  $d_t^{\text{int}}$  are the corresponding write decisions.

Given the interaction trajectory and feedback, the decoupler produces candidate explicit records, internalizable reasoning summaries, and write-or-skip decisions for the two substrates. This step is more than deciding whether a trajectory is worth keeping: a single interaction can contain both reusable factual records and reasoning experience, and these two signals should be written back through different mechanisms.

The decoupler therefore isolates these signals so that explicit-side content and implicit-side experience can be updated separately:

$$\mathcal{M}_{t+1}^{\text{ext}} = \begin{cases} \mathcal{U}_{\text{ext}}(\mathcal{M}_t^{\text{ext}}, E_t^{\text{ext}}), & d_t^{\text{ext}} = \text{write}, \\ \mathcal{M}_t^{\text{ext}}, & \text{otherwise}, \end{cases} \quad (10)$$

$$(\Delta_{t+1}, \mathcal{P}_{t+1}) = \begin{cases} \mathcal{U}_{\text{int}}(\Delta_t, \mathcal{P}_t, E_t^{\text{int}}, \tau_t, u_t), & d_t^{\text{int}} = \text{write}, \\ (\Delta_t, \mathcal{P}_t), & \text{otherwise}. \end{cases} \quad (11)$$

The two update operators implement different write mechanisms.  $\mathcal{U}_{\text{ext}}$  stores only the decoupled explicit-side signal so that reusable textual memory remains an editable external record.  $\mathcal{U}_{\text{int}}$  uses accepted reasoning experience to update the parameter weights and to append compact logic descriptors to the trajectory bank. Let the preserved implicit-memory samples up to time  $t$  be

$$\mathcal{B}_t = \{(E_i^{\text{int}}, u_i) \mid i \leq t, d_i^{\text{int}} = \text{write}\}, \quad (12)$$

then the implicit-memory update objective can be written as

$$\max_{\Delta} \mathbb{E}_{(E^{\text{int}}, u) \sim \mathcal{B}_t} [J_{\text{GRPO}}(\theta, \Delta; E^{\text{int}}, u)]. \quad (13)$$

For the trajectory bank, MoMem stores only the compact logic descriptor of the accepted trajectory

together with its anchoring request and metadata, so that future routing can recognize similar reasoning situations without replaying the full trajectory in the prompt.

By decoupling these updates, MoMem avoids two common failure modes of single-substrate memory. It prevents factual details and directly reusable procedural traces from being inappropriately compressed into parameter space, and it prevents task-specific or low-quality reasoning traces from being stored as long-term parameter memory. In this sense, the decoupler is what turns MoMem from a simple hybrid store into a memory system with differentiated learning dynamics.

## 4 Experiments

We evaluate MoMem in the setting that motivates the method: a lifelong agent must maintain explicit facts while also accumulating reusable reasoning and coding experience. In this section, complex logical experience in mathematics and coding is treated as implicit-memory content, while explicit memory is responsible for factual records and only those procedural traces that can be compactly verbalized. The experiments answer three questions: whether MoMem stays competitive on single-task benchmarks, whether it achieves better balance under mixed factual-implicit-experience streams, and which components are responsible for that robustness.

### 4.1 Experimental Protocol

**Task families.** We separate evaluation into factual memory and implicit experiential memory. Factual memory is tested on LongMemEval (Wu et al., 2024) and two factual settings from the official MemBench FirstAgent split (Tan et al., 2025): FirstAgent-Simple and FirstAgent-Conditional. Implicit experiential memory is tested on MATH-500 (Hendrycks et al., 2021; Lightman et al., 2023), Omni-MATH-500 (Gao et al., 2025), and Big-CodeBench (Zhuo et al., 2025), where success depends on reusable logical solution patterns rather than isolated facts or explicit workflow replay.

**Online evaluation.** All methods are evaluated as online memory systems. At round  $t$ , the model receives the current request, may read memory, answers, receives feedback when the benchmark provides it, and updates memory before round  $t+1$ . Memory is never reset within a stream, because the

goal is to test memory dynamics rather than one-shot retrieval.

**Mixed workloads.** To measure interference without introducing extra cross-task coupling, we construct three pairwise mixed streams, each containing one factual setting and one implicit-experience setting: LongMemEval + MATH-500, MemBench FirstAgent-Simple + Omni-MATH-500, and MemBench FirstAgent-Conditional + BigCodeBench. The two task types are interleaved online with a fixed shared order, and memory is never reset.

For compact presentation, we abbreviate only the two long MemBench factual settings as MB-S and MB-C.

**Baselines.** We compare against baselines organized by both memory function and storage form. The text-based factual-memory group includes Mem0, MemoryBank, MemGPT, and A-Mem (Chhikara et al., 2025; Zhong et al., 2024; Packer et al., 2023; Xu et al., 2025). The explicit workflow/trace-based experience group includes AWM, Dynamic-Cheatsheet, and ReasoningBank (Wang et al., 2024b; Suzgun et al., 2025; Guan et al., 2025). The parametric factual-memory baseline is SFT, and the implicit experiential-memory baseline is GRPO. This setup separates textual versus parametric storage, and factual records versus complex logical experience.

**Metrics.** All evaluation instances have reference answers or executable tests, so we report accuracy on a 0–100 scale throughout. For factual-memory benchmarks, accuracy is measured by answer correctness under the benchmark evaluator. For mathematical reasoning, we report exact-match final-answer accuracy. For coding, we report unit-test pass accuracy under the benchmark’s official harness. In single-task evaluation, we macro-average LongMemEval, MemBench FirstAgent-Simple, and MemBench FirstAgent-Conditional into  $S_F$ , and macro-average MATH-500, Omni-MATH-500, and BigCodeBench into  $S_E$ . In mixed evaluation, each stream contains one factual benchmark and one implicit-experience benchmark, so we report the corresponding pairwise accuracies directly. This avoids assigning an artificial weight between factual and implicit-experience tasks.

**Implementation details.** All methods use the same backbone family and answer-generation prompt unless a baseline requires its original format. Retrieval-based methods use matched re-

trieval budgets and context limits, and parameter-update methods use matched LoRA capacity and update schedules. For MoMem, the router and decoupler use structured outputs with deterministic decoding. Hyperparameters are selected on development streams and fixed before test evaluation.

## 4.2 Main Results

**Single-task evaluation.** We first evaluate each method on the six settings separately. This is a specialization check rather than the paper’s main claim: a hybrid method need not dominate every specialist, but it should avoid clear deficits on either factual retention or implicit reuse of complex logical experience. Table 1 reports this comparison. Table 1 is mainly about the shape of each method’s strengths. MoMem is successful here if it remains competitive across factual, math, and coding settings rather than collapsing outside one preferred regime.

**Mixed-stream evaluation.** Table 2 reports the central experiment. Each block evaluates one pairwise stream containing exactly one factual benchmark and one implicit-experience benchmark, which exposes interference that single-task evaluation can miss.

Table 2 should be read along two axes: factual retention and implicit reuse of complex logical experience. The key mixed-workload criterion is not high accuracy on only one side, but retaining both task-family accuracies once both demands share the same evolving memory state.

## 4.3 Ablation Study

We ablate the mechanisms that coordinate the two memory substrates on the same three mixed settings. *w/o Router* removes query-dependent invocation, *w/o Decoupler* removes post-interaction signal separation, and *w/o Router & Decoupler* removes both control points. Table 3 reports these comparisons.

The ablations are diagnostic. If the router matters, removing it should mainly reduce one or both task-family accuracies by causing unnecessary prompt injection or implicit activation. If the decoupler matters, removing it should reduce accuracy because factual signals and complex logical experience contaminate each other’s update path. The combined ablation tests whether hybridization alone is sufficient.

| Method             | LongMem Eval | MB-S        | MB-C        | Factual Acc. Avg. | MATH -500   | Omni-MATH -500 | BigCode Bench | Implicit Exp. Avg. |
|--------------------|--------------|-------------|-------------|-------------------|-------------|----------------|---------------|--------------------|
| Mem0               | 71.8         | 74.2        | 73.4        | 73.1              | 33.0        | 41.0           | 24.2          | 32.7               |
| MemoryBank         | 69.5         | 70.1        | 70.2        | 69.9              | 31.4        | 39.2           | 22.8          | 31.1               |
| MemGPT             | 68.7         | 68.8        | 68.9        | 68.8              | 30.1        | 37.5           | 21.7          | 29.8               |
| A-Mem              | 74.6         | 75.0        | 75.1        | 74.9              | 34.5        | 42.4           | 25.5          | 34.1               |
| AWM                | 58.6         | 60.1        | 61.3        | 60.0              | 42.1        | 51.8           | 33.7          | 42.5               |
| Dynamic-Cheatsheet | 56.9         | 58.4        | 59.8        | 58.4              | 45.6        | 54.0           | 35.9          | 45.2               |
| ReasoningBank      | 55.2         | 56.5        | 58.0        | 56.6              | 49.8        | 58.6           | 38.1          | 48.8               |
| SFT                | 66.2         | 66.8        | 67.6        | 66.9              | 35.7        | 42.9           | 27.8          | 35.5               |
| GRPO               | 52.4         | 53.9        | 55.1        | 53.8              | 51.6        | 60.5           | 40.7          | 50.9               |
| <b>MoMem</b>       | <b>76.8</b>  | <b>77.4</b> | <b>78.2</b> | <b>77.5</b>       | <b>53.4</b> | <b>62.4</b>    | <b>42.3</b>   | <b>52.7</b>        |

Table 1: Single-task answer accuracy on the six settings. Numbers are illustrative placeholders for layout only and must be replaced with measured results.

## 5 Discussion

MoMem is motivated by a simple observation: long-term memory in agents is not monolithic. If a lifelong agent must preserve rapidly changing explicit facts while also accumulating reusable strategies, its memory architecture should reflect that heterogeneity. MoMem therefore treats memory not as a single store, but as a coordinated system in which different substrates have different responsibilities, invocation rules, and update mechanisms.

This perspective also suggests a broader evaluation principle. Beyond asking whether a memory system improves downstream accuracy on isolated benchmarks, we should ask whether it remains robust once different memory demands are repeatedly interleaved. The mixed settings in our experiments are intended as controlled diagnostic proxies for this question, where robustness to cross-task trade-offs is treated as a first-class objective.

## 6 Conclusion

We present MoMem, a hybrid memory architecture for lifelong LLM agents that decouples factual memory from experiential memory across storage, invocation, and update. MoMem combines explicit memory for precise and editable factual knowledge, implicit memory for transferable strategic experience, a router for query-dependent memory activation, and a decoupler for post-interaction memory separation. Rather than framing success as universal dominance on every isolated benchmark, we argue that agent memory systems should also be judged by how robustly they handle heterogeneous workloads. We therefore advocate mixed cross-task evaluation as a diagnostic complement to standard isolated benchmarks when studying long-term memory in realistic agents.

## Limitations

MoMem introduces additional system complexity relative to single-carrier memory architectures. Its effectiveness depends on the reliability of the router and decoupler, as well as the quality of the feedback signal used to update implicit memory. The implicit module may also inherit the stability challenges of continual parameter updating, while the explicit module raises privacy and storage concerns when factual memory contains user-specific information. Finally, although mixed-memory evaluation better reflects realistic workloads, benchmark construction choices, such as the interleaving schedule and task ratio, may materially affect empirical conclusions.

## Acknowledgments

Acknowledgments are omitted in the review version.

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| Setting                | Method             | Factual Acc. | Implicit Exp. Acc. |
|------------------------|--------------------|--------------|--------------------|
| LongMemEval + MATH-500 | Mem0               | 67.2         | 31.4               |
|                        | MemoryBank         | 64.1         | 29.8               |
|                        | MemGPT             | 62.8         | 28.5               |
|                        | A-Mem              | 70.4         | 33.1               |
|                        | AWM                | 51.0         | 39.7               |
|                        | Dynamic-Cheatsheet | 49.6         | 43.3               |
|                        | ReasoningBank      | 48.2         | 47.5               |
|                        | SFT                | 60.1         | 33.5               |
|                        | GRPO               | 45.8         | 49.2               |
|                        | <b>MoMem</b>       | <b>74.6</b>  | <b>51.8</b>        |
| MB-S + Omni-MATH-500   | Mem0               | 69.4         | 39.2               |
|                        | MemoryBank         | 65.8         | 37.6               |
|                        | MemGPT             | 64.0         | 35.9               |
|                        | A-Mem              | 71.1         | 40.8               |
|                        | AWM                | 56.6         | 49.1               |
|                        | Dynamic-Cheatsheet | 54.8         | 51.5               |
|                        | ReasoningBank      | 53.5         | 55.9               |
|                        | SFT                | 62.7         | 40.5               |
|                        | GRPO               | 48.1         | 57.4               |
|                        | <b>MoMem</b>       | <b>75.8</b>  | <b>58.7</b>        |
| MB-C + BigCodeBench    | Mem0               | 68.3         | 22.1               |
|                        | MemoryBank         | 65.0         | 20.6               |
|                        | MemGPT             | 63.8         | 19.4               |
|                        | A-Mem              | 70.9         | 23.5               |
|                        | AWM                | 52.0         | 30.2               |
|                        | Dynamic-Cheatsheet | 50.1         | 33.0               |
|                        | ReasoningBank      | 48.5         | 35.1               |
|                        | SFT                | 58.4         | 26.0               |
|                        | GRPO               | 44.6         | 37.8               |
|                        | <b>MoMem</b>       | <b>72.4</b>  | <b>39.6</b>        |

Table 2: Main answer-accuracy results on three pairwise factual–implicit–experience mixed streams. Numbers are illustrative placeholders for layout only and must be replaced with measured results.

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| Setting                   | Variant                | Factual Acc. | Implicit Exp. Acc. |
|---------------------------|------------------------|--------------|--------------------|
| LongMemEval<br>+ MATH-500 | MoMem                  | <b>74.6</b>  | <b>51.8</b>        |
|                           | w/o Router             | 70.2         | 47.9               |
|                           | w/o Decoupler          | 67.1         | 48.8               |
|                           | w/o Router & Decoupler | 63.5         | 43.2               |
| MB-S<br>+ Omni-MATH-500   | MoMem                  | <b>75.8</b>  | <b>58.7</b>        |
|                           | w/o Router             | 71.5         | 54.1               |
|                           | w/o Decoupler          | 68.6         | 55.0               |
|                           | w/o Router & Decoupler | 64.9         | 49.5               |
| MB-C<br>+ BigCodeBench    | MoMem                  | <b>72.4</b>  | <b>39.6</b>        |
|                           | w/o Router             | 67.8         | 35.0               |
|                           | w/o Decoupler          | 64.9         | 35.4               |
|                           | w/o Router & Decoupler | 61.0         | 31.7               |

Table 3: Ablation answer accuracy on the three pairwise mixed settings. Numbers are illustrative placeholders for layout only and must be replaced with measured results.

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